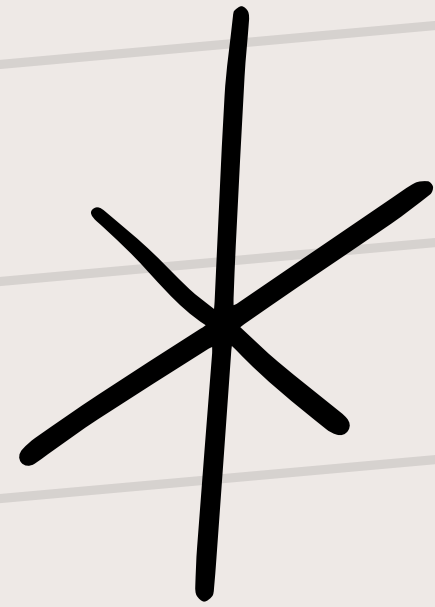
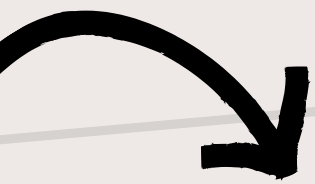




Project Proposal



Group 9





Why Accent Robustness in Speech Translation

Speech-to-text translation systems must process English spoken with **diverse regional accents**.

Accent variation can substantially affect speech recognition performance, especially for underrepresented accents (*Prabhu et al., 2023*)

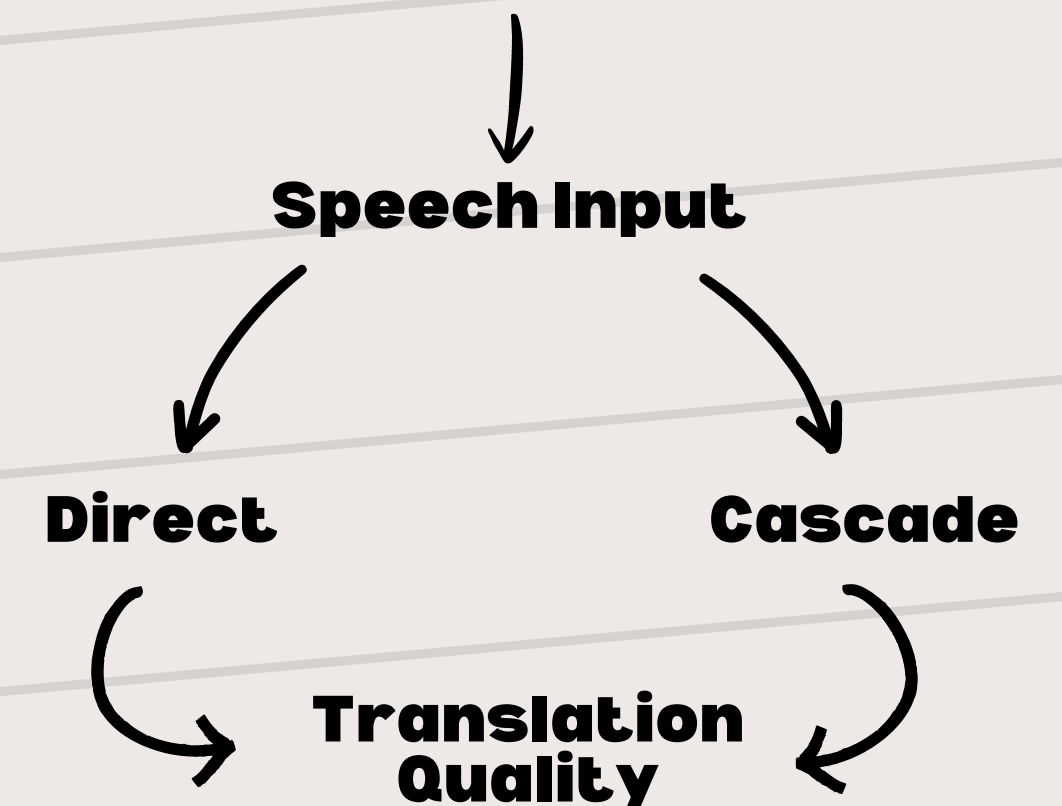
In a **Cascaded** system, ASR errors can propagate into the downstream translation stage

A **Direct** system translate speech without an explicit intermediate transcript, so it may respond differently to accent variation

Open question...

Do regional English accents affect final translation quality differently in Direct and Cascaded systems?

Regional English Accent



Why it matters

- Accent-sensitive performance can reduce the reliability and inclusiveness of speech-translation systems
- Accent robustness in ASR and Direct vs. Cascaded ST have been studied, but not combined



Research Question

How does regional English accent affect the translation quality of direct and cascaded speech-to-text translation systems?



1

Construct a balanced evaluation set

5 regional accent groups x 600 clips each $\rightarrow$ N = 3,000

2

Compare two ST systems on the same speech

Cascade: Whisper $\rightarrow$ NLLB | Direct: SeamlessM4T v2

3

Measure translation quality across accent groups

chrF++ (primary), BLEU (secondary)

4

Measure accent-related ASR variation

Whisper WER vs. Common Voice reference transcript

5

Link ASR errors to Cascade translation degradation

Association analysis, not causal claim



Expected Contribution

A systematic comparison of how regional English accent groups are associated with translation performance in representative Direct and Cascaded ST systems.



Robinson et al. (2024): Dialectal and Low-resource System Description

What they Investigated

Compared modern pretrained model families for dialectal and low resource speech translation across three system types:

Direct	Speech → SeamlessM4T v2 → Translation
Direct	Speech → Whisper → Translation
Cascade	Speech → Whisper ASR → NLLB MT → Translation

Key finding

- No single best approach for all language pair.
- SeamlessM4T v2 performed best for half of the evaluated directions; cascaded approaches perform better for the rest.

What we take

Robinson et al.	Limitation	Our extension
Direct vs Cascade	Not english accent focused	Direct vs Cascade across English accent
SeamlessM4T v2	Different language pairs	Same source language (English)
Whisper → NLLB	Fine-tuned systems	Frozen pretrained evaluation

Prabhu et al. (2023): Accented Speech Recognition With Accent-specific Codebooks

What they Investigated

- Accented English ASR using Mozilla Common Voice.
- 14 accent groups: 5 seen, 9 unseen during training.
- Speaker-disjoint train/develop/test design.

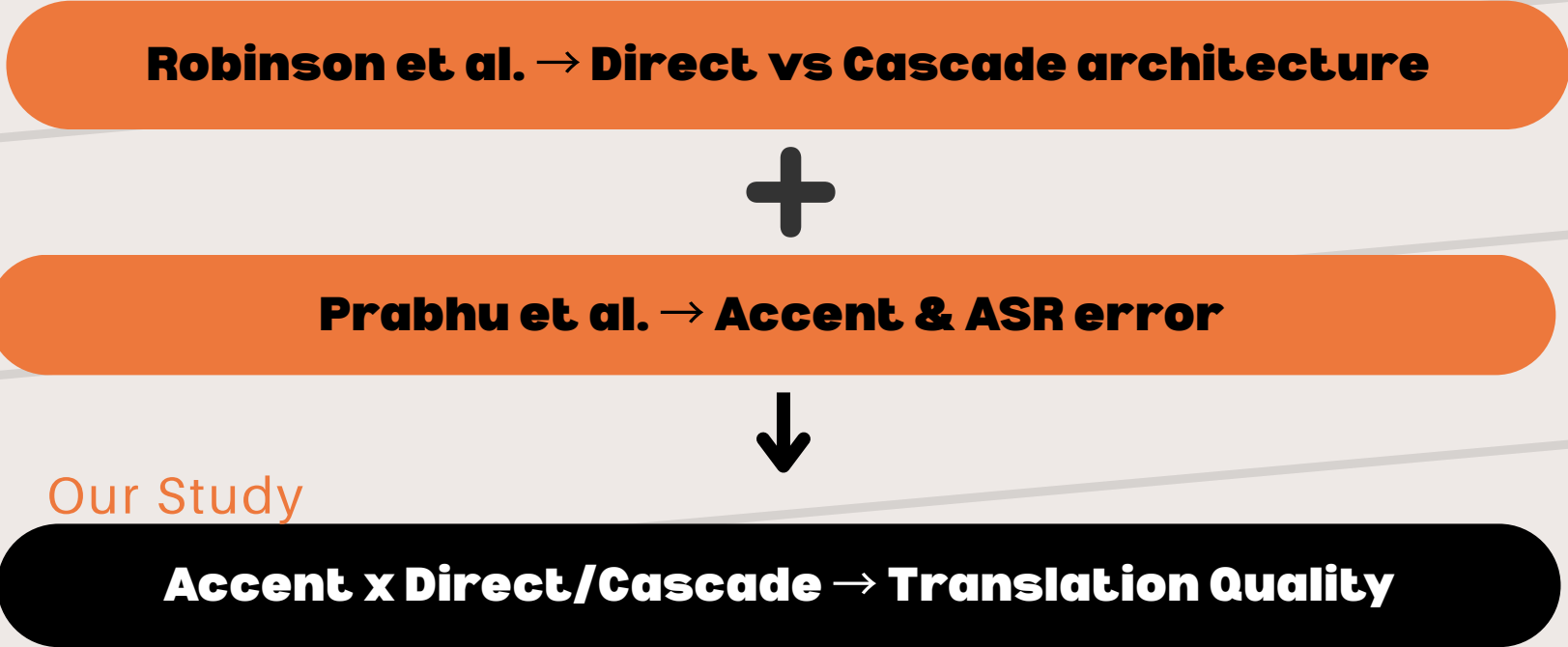
Key finding

- Accent sensitivity:
 - WER of 14% for seen accent
 - WER of 23.7% for unseen accent

What we take

Prabhu et al.	Limitation	Our extension
Accent → ASR (WER only)	Stop at transcription, no translation	Accent → Speech Translation quality
Seen vs unseen accent split	Built for adaptation, not architecture comparison	Compare Direct vs Cascade within accent group

Combining the two



Common Voice + CoVoST2: A Balanced, Reference-Verified Evaluation Set

Mozilla Common Voice (English)
(Ardila et al., 2020)
English speech recordings with source transcripts and regional accent labels
+
CoVoST2
(Wang et al., 2021)
human-translated references for a subset of Common Voice sentences (English to Chinese)

Filter by:

- 1. Primary accent tag
- 2. CoVoST2 textmatch with verified Chinese translation
- 3. 1st-99th percentile audio duration
- 4. Speaker cap (max 100 clips/speaker)

Selected accent groups (post cap)

<u>Accent</u>	<u>Available</u>
India & South Asia	24096
Southern African	4422
Filipino	1817
Malaysian English	1204
Hong Kong English	<u>695</u>

Sample 600 clips per accent group (limit speaker dominance, keep the groups balanced for evaluation)

Set N:

Total N = 3000 (After Sampling)

<u>Accent</u>	<u>Sample</u>
India & South Asia	600
Southern African	600
Filipino	600
Malaysian English	600
Hong Kong English	600



Limitation

Sentence overlap falls off fast! 4110 sentences are shared by at least two accent groups, but only 6 across all five, so we sample each group independently, with duration and speaker controls to keep it fair



Reference quality

Every clip in our core evaluation set is matched to a real human translation from CoVoST2, so our scoring never depends on NLLB-generated references.

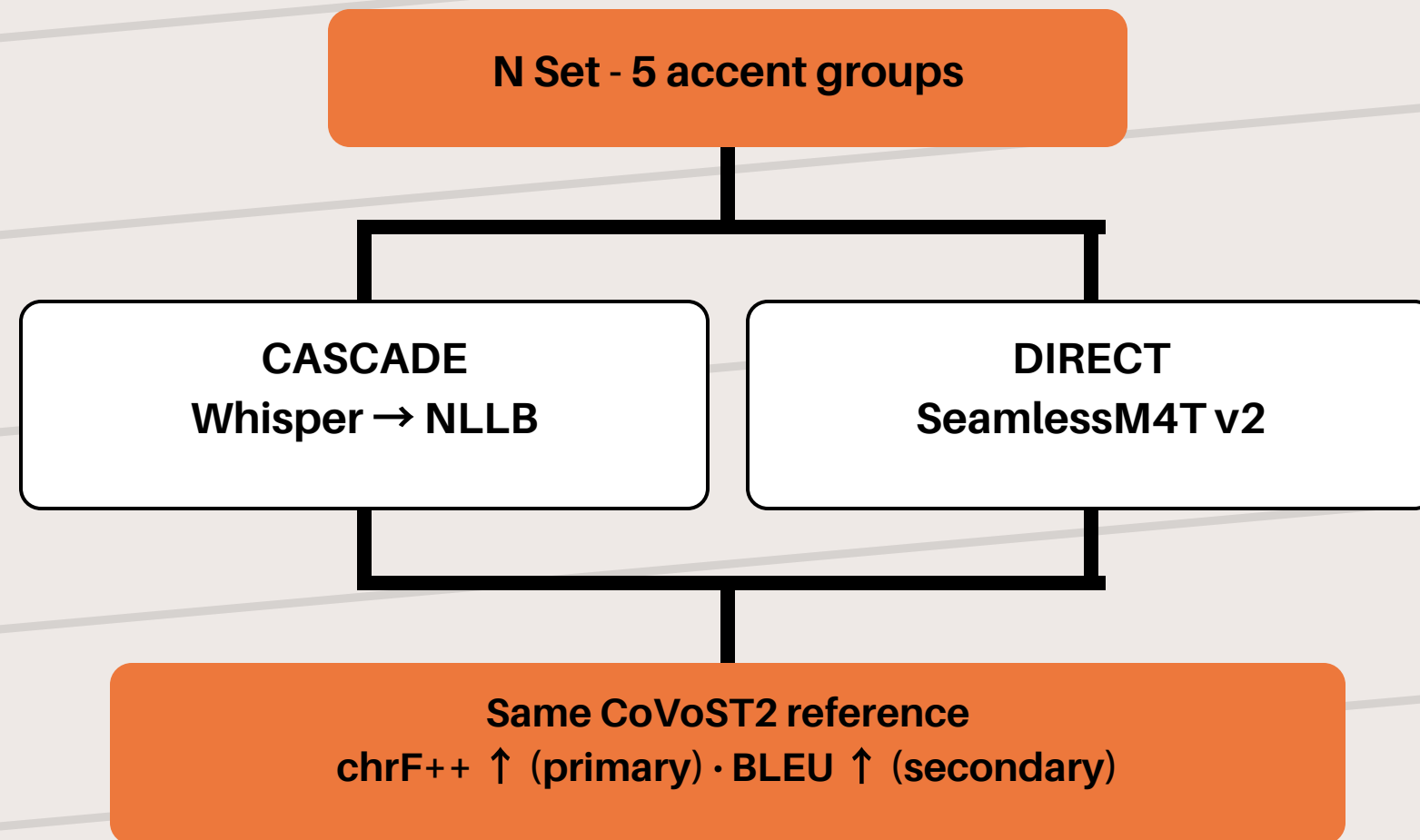


Data Sources: 1) [Common Voice Scripted Speech 26.0 - English](#); 2) [CoVoST: A Large-Scale Multilingual Speech-To-Text Translation Corpus](#)

Colab Link: [Dataset verification notebook](#)

In-slide Citations: 1) [Ardila et al. \(2020\) Common Voice: A Massively-Multilingual Speech Corpus](#); 2) [Wang et al. \(2021\) CoVoST 2 and Massively Multilingual Speech Translation](#)

Evaluation



WER ↓ (Cascade only): Whisper transcript vs Common Voice — error-propagation diagnostic. Paired design: any per-clip difference is due to the system, not the sentence (Bentivogli et al., 2021).

Metrics at a glance — and why

chrF++ ↑ — PRIMARY (Popović, 2017)

Character + word n-gram overlap with the human reference. Fair on near-misses and robust when the two systems tokenise differently.

BLEU ↑ — SECONDARY (Papineni et al., 2002)

Word n-gram precision. Kept so our results compare directly with prior speech-translation work (Robinson et al., 2024).

WER ↓ — DIAGNOSTIC, Cascade only

Whisper transcript vs Common Voice transcript. Shows where recognition errors enter and propagate into the translation.

Metrics - Detailed



Metric	What it grades	Example	Better
 chrF++ (primary) <i>Popović, 2017</i> Character n-gram F-score with word order. Gives partial credit for near-misses.	Final translation (both systems)	Reference: "running" Output: "runs" BLEU gives 0 credit (word mismatch); chrF++ gives partial credit for the shared r-u-n.	 Higher ↑
 BLEU (secondary) <i>Papineni et al., 2002</i> Precision of matching n-grams with brevity penalty.	Final translation (both systems)	Reference: "the weather is very cold today" Output: "the weather is cold today" Matching word chunks counted; missing "very" is penalised.	 Higher ↑
 WER (diagnostic) <i>Cascade only</i> Word Error Rate between Whisper transcript and Common Voice reference.	Whisper transcript vs Common Voice	Reference: "I need a sheet of paper" Whisper: "I need a seat of paper" 1 error / 6 words = 16.7% — $WER = (S + D + I) / N$.	 Lower ↓



Key takeaway

We use chrF++ as our primary translation metric, BLEU as a complementary check, and WER to diagnose ASR (Whisper) quality in the Cascade system.

STATISTICAL ANALYSIS

Three analyses to identify whether accent affects the two systems differently — and where the difference may originate.

01 PERFORMANCE GAP

Question: Does accent affect Direct and Cascade differently?

BRIEF EXPLANATION

For each accent group, calculate:

$$\text{Gap} = \text{Direct chrF++} - \text{Cascade chrF++}$$

ILLUSTRATIVE OUTPUT

Accent Group	Direct chrF++	Cascade chrF++	Gap (Direct – Cascade)
Accent A	52.0	48.0	+4.0
Accent B	51.0	42.0	+9.0

WHAT WOULD THIS MEAN?

If the gap changes substantially across accents, one system is more accent-sensitive than the other.

Similar gaps across accents
→ similar relative sensitivity

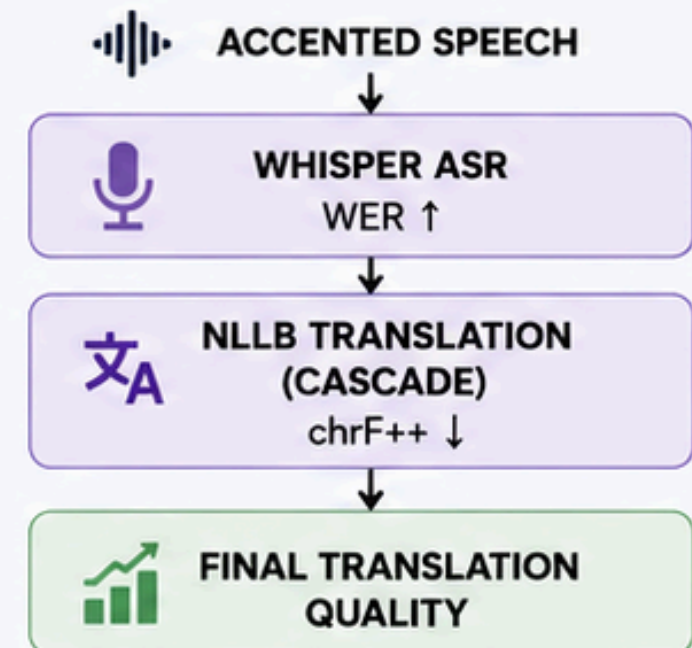
Varying gaps across accents
→ differential accent sensitivity

02 CORRELATION

Question: Is Cascade degradation associated with ASR errors?

BRIEF EXPLANATION

Compute Spearman correlation between **Whisper WER** and **Cascade chrF++** across clips.



STRONG NEGATIVE CORRELATION ($\rho < 0$)
WER ↑ → Cascade quality ↓
Evidence that ASR errors may be propagating downstream.

WEAK OR NO CORRELATION
Cascade degradation may involve other factors, including the downstream MT stage.

Diagnostic association, not causal proof.

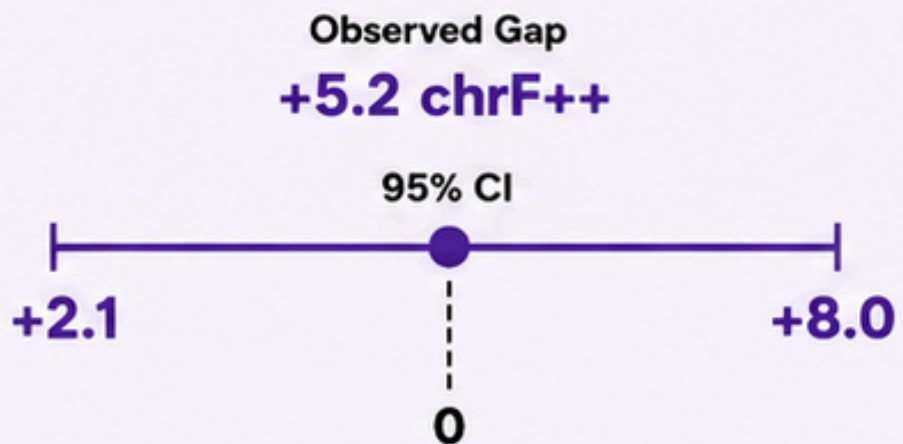
03 PAIRED BOOTSTRAP

Are the observed Direct–Cascade differences reliable?

WHAT IS IT?
Repeatedly resample the same evaluated clips (with replacement) and recalculate the Direct–Cascade performance gap.

WHY PAIRED?
Every clip is processed by both systems, so their scores form a natural pair. This controls for item-level variability.

BENEFIT
Produces a 95% confidence interval (CI) around the performance gap to judge statistical reliability.



✓ CI excludes 0 → evidence of a reliable Direct–Cascade difference

✗ CI includes 0 → difference is not clearly distinguishable from zero

Planned Results Layout

Gap = Direct chrF++ – Cascade chrF++.

Accent group	WER ↓	Direct chrF++ ↑	Cascade chrF++ ↑	Gap (D – C)
India & South Asia	—	—	—	—
Southern African	—	—	—	—
Filipino	—	—	—	—
Malaysian English	—	—	—	—
Hong Kong English	—	—	—	—

Constant gap across groups ⇒ equal accent sensitivity · Varying gap ⇒ one architecture is more accent-sensitive (accent × architecture interaction ; core of our RQ)

Paired Evaluation of Cascade and Direct

Baselines

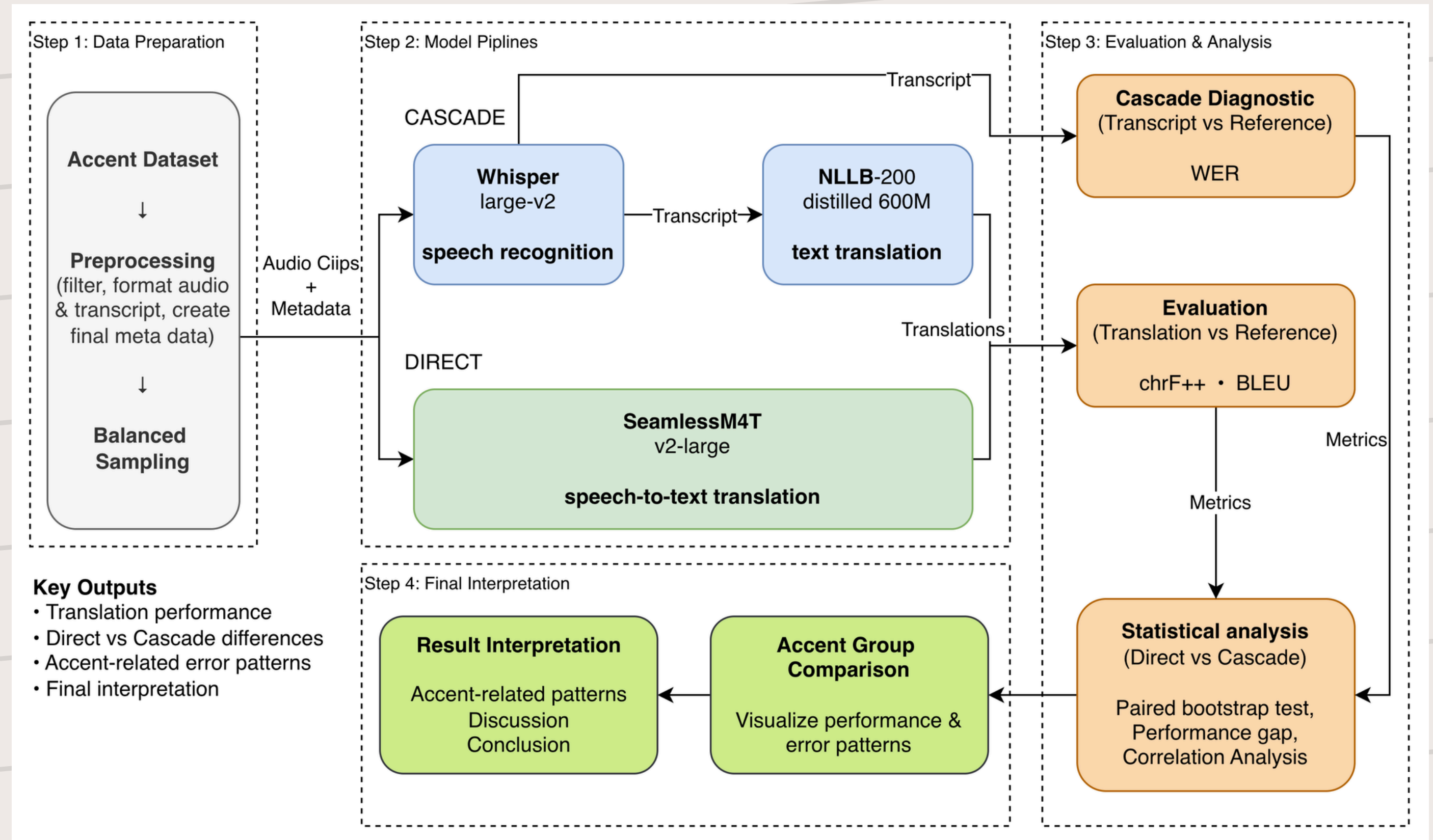
Reference systems adopted from Robinson et al. (2024):

- Direct: SeamlessM4T
- Cascade: Whisper → NLLB

Contribution: Benchmark established Direct and Cascaded ST systems to evaluate accent sensitivity, not to develop a new model.

Controlled Experiment

- Same audio clips
- Same target/reference set
- No accent-specific adaptation
- No fine-tuning initially



COMPSCI 760 | Feasibility,
Timeline, Risks & Team

FEASIBILITY

3,000 CLIPS TOTAL

- 5 accent groups
- 600 clips per group
- Balanced sampling

PRETRAINED SYSTEMS

- Direct: SeamlessM4T v2-large
- Cascade: Whisper large-v2 → NLLB-200 distilled 600M

NO MODEL FINE-TUNING

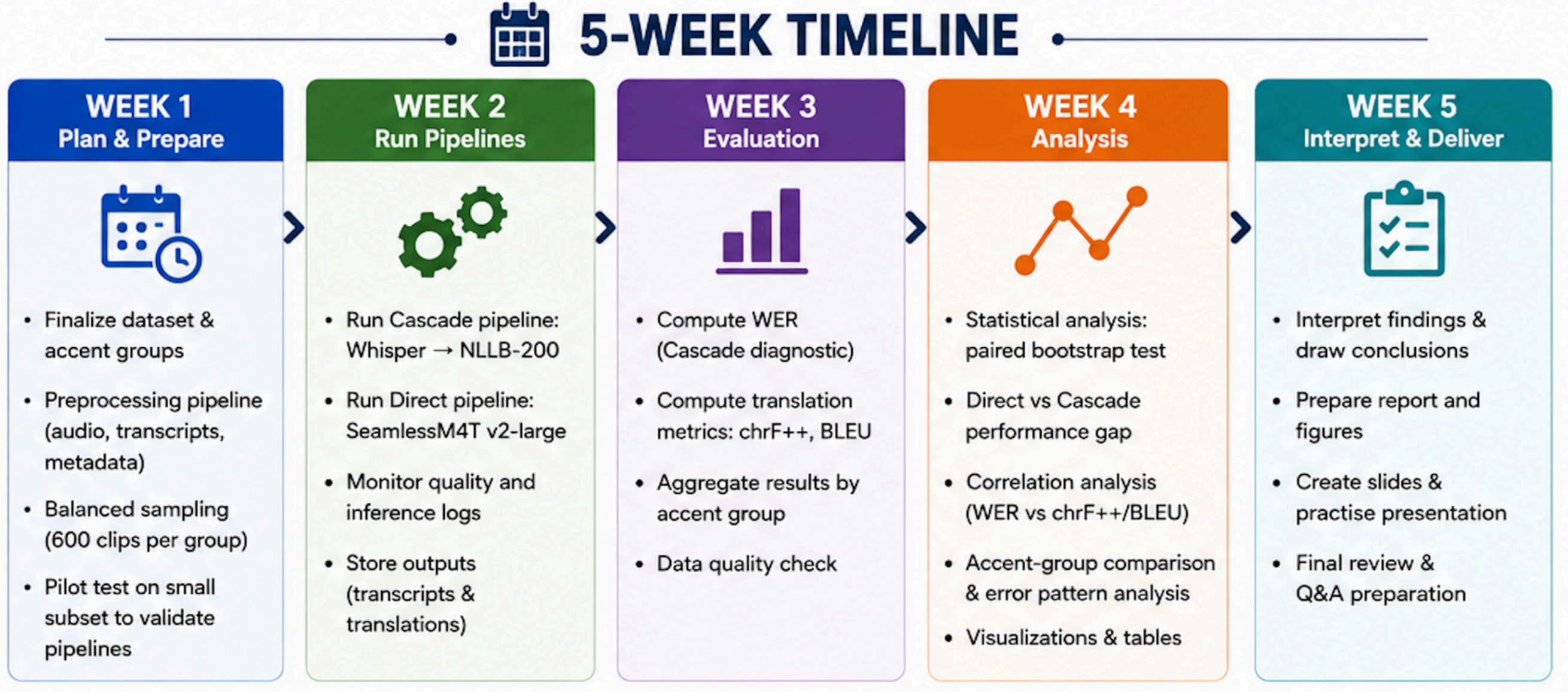
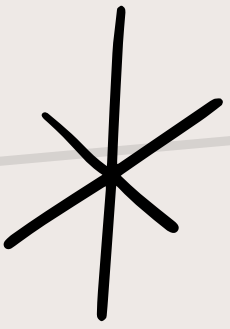
We use existing pre-trained models as-is.

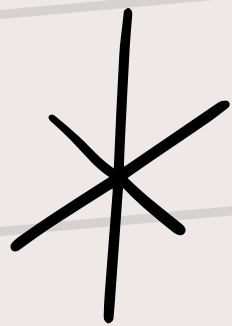
EXISTING REFERENCE TRANSCRIPTS AND TRANSLATIONS

Used for evaluation (WER, chrF++, BLEU).

PARALLEL TEAM WORKFLOW

Tasks run in parallel across 5 team members, reducing total project time.





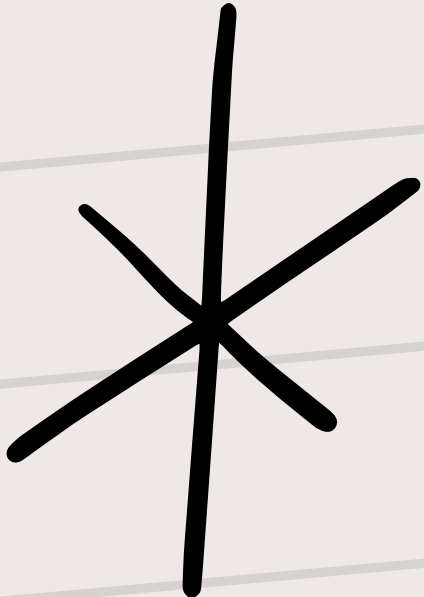
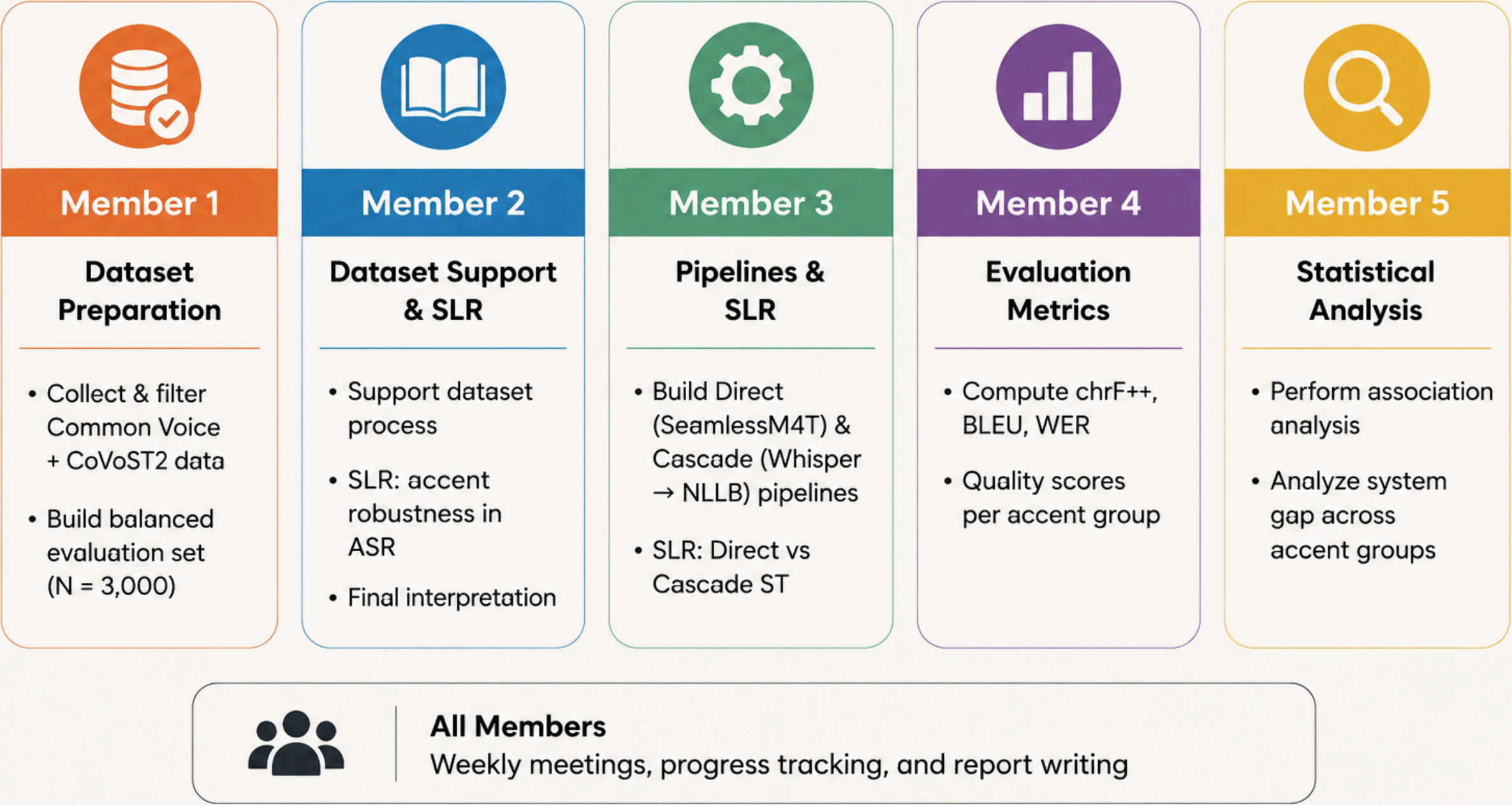
RISKS & PLAN B

RISK	POTENTIAL IMPACT	MITIGATION / PLAN B
 Insufficient usable samples after filtering	May not reach target (600 clips per accent group)	 Lower target equally across all groups, keep groups balanced, and extend duration if needed.
 Dataset/content imbalance across accents	Biased results and unfair comparisons	 Balance by duration, transcript length, and speaker contribution, and report any remaining imbalance transparently.
 High inference cost / long processing time	Delays in completing experiments	 Process in parallel or in batches on available GPUs, reduce optional analyses if needed, and prioritize the core experiment first.
 Model or target-language compatibility issue	Translation failures or low-quality outputs	 Validate with a small pilot set first and use an alternative target language/model if needed.
 Unbalanced contributions from team members	Uneven workload and lower overall quality	 Define clear tasks and weekly goals, use progress check-ins to monitor contributions, and redistribute work early if needed.

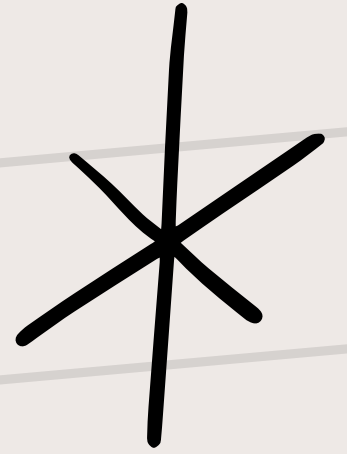


Member 1: Arizona
Member 2: Zhitong
Member 3: Nathan
Member 4: Patricia
Member 5: Jidini

* Team Roles & Responsibilities



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GenAI Use

ChatGPT was used for brainstorming, research support, wording refinement, and generating slide visuals.

All experimental decisions, analysis, results, and conclusions are the group's own work.



**Thank
You**

